

Code-Layout, Readability and Reusability

Date	5 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained in HTML, CSS, JavaScript, and Python files.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, models, and database.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names improve readability.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as <code>predict_crop()</code> and <code>login_user()</code> clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added for important sections of the code.
6	Modular Design	Code is split into reusable functions/modules	Yes	Backend logic, ML model, and frontend are organized into separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated code is minimized through reusable functions.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are

				implemented in the backend
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Login Module	Python (Flask)	User Authentication	High
2	Crop Prediction Module	Python, Scikit-learn	Crop Recommendation	High
3	Database module	SQLite	User Data & Prediction History	High
4	User Interface Components	HTML, CSS, JavaScript, Bootstrap	Multiple Web Pages	Medium
5	Machine Learning Model	Scikit-learn, Joblib	Prediction Service	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate frontend and backend.
Readability	5	Clean code with meaningful names and consistent formatting.
Reusability	5	Modules and functions are reusable across the application.
Documentation / Comments	4	Important code sections are documented with comments.
Overall Score	4.8/5	The project follows good coding standards and maintainable design practices.